De Broglie's Hypothesis

Objectives:

- Describe de Broglie's hypothesis of matter waves
- Explain how the de Broglie's hypothesis gives the rationale for the quantization of angular momentum in Bohr's quantum theory of the hydrogen atom
- Describe the Davisson–Germer experiment

- An electromagnetic wave can behave like a particle of light when interacting with matter as in Compton's formula
- Louis de Broglie proposed a new speculative hypothesis that electrons and other particles of matter can behave like waves
- This idea is known as de Broglie's hypothesis of matter waves
- De Broglie's hypothesis, together with Bohr's early quantum theory, led to the development of a new theory of wave quantum mechanics to describe the physics of atoms and subatomic particles.

Quantum mechanics has paved the way for new engineering inventions and technologies, such as the laser and magnetic resonance imaging (MRI). These new technologies drive discoveries in other sciences such as biology and chemistry.

According to de Broglie's hypothesis, massless photons as well as massive particles must satisfy one common set of relations that connect the energy *E* with the frequency v and the linear momentum *p* with the wavelength *λ*

Any particle that has energy and momentum is a de Broglie wave of frequency v and wavelength λ

$$E = hv \qquad (1)$$

$$\lambda = \frac{h}{p} \qquad (2)$$

***E, p* and h are, respectively, the relativistic energy, the momentum of a particle and the Planck's constant.**

De Broglie's relations are usually expressed in terms of the

wave vector $\vec{k}$, $k = 2\pi/\lambda$

and the wave frequency $\omega = 2\pi\nu$

as done usually for waves.:

Equations 1 and 2 become:

$$E = \hbar\omega \qquad (3)$$

$$\vec{p} = \hbar\vec{k} \qquad (4)$$

From the wave theory, a wave carries its energy with the group velocity.

For matter waves, this group velocity is the velocity u of the particle.

The relativistic energy of a particle is given by:

$$E = mc^2 \quad (5)$$

The relativistic momentum of a particle is given by:

$$p = mu \quad (6)$$

Hence de Broglie relations for matter waves satisfy the following relation from the above equations:

$$\lambda v = \frac{\omega}{k} = \frac{E/\hbar}{p/\hbar} = \frac{E}{p} = \frac{mc^2}{mu} = \frac{c^2}{u} = \frac{c}{\beta} \quad (7)$$

where $\beta = u/c$

When a particle is massless we have $u = c$ and Equation 7 becomes

$$\lambda v = c \tag{8}$$

Strategies for solving problems here:
Use Eq. 2 to find the de Broglie wavelength.
When the problem involves a *nonrelativistic* object moving with a *nonrelativistic speed u*, such that $\beta = u/c \ll 1$, we use *nonrelativistic momentum p*.
When the *nonrelativistic approximation* cannot be used, we must use the relativistic momentum

$$p = mu = m_0 \gamma \beta / c \tag{9}$$

where the rest mass energy of a particle is $E_0 = m_0c^2$ and γ is the Lorentz factor $\gamma = 1/\sqrt{1-\beta^2}$

The total energy E of a particle is given by Eq. 1 and the kinetic energy is

$$K = E - E_0 = (\gamma - 1)E_0 \qquad (10)$$

According to the theory of special relativity, any particle in nature obeys the relativistic energy equation

$$E^2 = p^2c^2 + m_0^2c^4 \qquad (11)$$

Where E is the total energy of a particle,
P is its linear momentum, and
m_0 is its rest mass.
When the kinetic energy is known, we can find the momentum from Eq. 1:

$$p = \sqrt{(E^2 - E_0^2)/c^2} = \sqrt{K(K + 2E_0}/c \qquad (12)$$

Using Eq. 12 in Eq. 2 to get

$$\lambda = \frac{h}{p} = \frac{hc}{\sqrt{K(K + 2E_0}} \qquad (13)$$

CLASS WORK

Calculate the de Broglie wavelength of:

1. a 0.65-kg basketball thrown at a speed of 10 m/s,

2. a nonrelativistic electron with a kinetic energy of 1.0 eV, and

3. a relativistic electron with a kinetic energy of 108 keV.